

Report

*Design of an autonomous mobile cleaning
robot*

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Date of Submission: 01.08.2025

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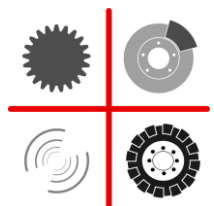


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0 Notation and Abbreviation

Abbreviation/Symbol	SI-Units	Terminology
AMR		Autonomous Mobile Robot

1 Introduction

With the rapid advancements in the field of autonomous systems and robotics, Autonomous Mobile Robots (AMRs) have been widely developed and utilized in various sectors, particularly for repetitive, labor-intensive and time-consuming tasks. Among these, automated cleaning robots have gained a lot of attention in the recent years due to their potential to reduce labor dependency, improve consistency and increase operational efficiency

Manual cleaning in laboratory environments like the test hall of LMS (Labor für Maschinensystem) can be inefficient, resource- and time intensive as well as difficult to maintain consistently. Furthermore, the plan, workspace and equipment and large machinery filled environment of the LMS test hall complicates precise and consistent manual cleaning. Thus, an autonomous cleaning robot which can autonomously and efficiently clean the test hall would be an optimal solution.

The goal of this project is to design an autonomous cleaning robot capable of moist cleaning the LMS test hall. The robot should operate fully autonomously, cleaning the entire designated area which conforming to applicable technical and safety standards for autonomous machines.

The robot is expected to:

- Increase cleaning efficiency
- Reduce operational and labor cost
- Minimize human intervention for repetitive cleaning task
- Ensure clean working environment consistently

This project encompasses the designing process from the requirement analysis to conceptual design, route planning, and system integration, including:

- Analysis and determination of technical, functional and operational requirements for the robot
- Identification and analysis of the safety standards
- Definition of system specifications
- Development of basic concepts and assemblies
- Route planning and movement strategies
- Documentation

Through these steps, the project aims to deliver a conceptual design of autonomous cleaning robot that meets all the technical, functional and operational demands for autonomous cleaning of LMS test hall while complying with the safety standards and exceeding performance expectations.

2 Analysis and Determination of Requirements

This section focuses on identifying, analyzing and determining the requirements for the autonomous cleaning robot to operate effectively within LMS test hall, based on demand and wishes regarding the functionality and safety standard needed. The technical, functional and operational aspect of the robot as well as the environment, nature of cleaning task and expectation of end user will be analyzed. These requirements form the foundation of conceptual design and guide all subsequent development phase. Additionally, special emphasis will be place on safety considerations ensuring the robot complies with relevant standard for autonomous machines operating indoor in human -occupied laboratory environment.

2.1 Analysis of requirements

To ensure a successful design and operation of autonomous cleaning robot, it is essential to thoroughly analyze the operating environment, nature of cleaning task and user expectation regarding the robot. This analysis serves as basis for determining the functional, technical, operational and safety requirements.

2.1.1 Operational Environment

The robot will operate within the LMS test hall which is indoor environment typically used for development and testing of mechanical system in TU berlin. The hall comprises of flat surface presumably concrete floor. It is filled with heavy machinery, fixed installations, equipment, cables. In addition to them, often researchers are working in the LMS test hall, hence the robot should be capable of autonomously navigating around while avoiding obstacle and collision as well as be safe for human and delicate lab equipment. As the robot is designed to moist clean, special attention needs to be given to avoid excessive wetting or water usage by robot to prevent damage of lab equipment or health hazards which can be cause by slippery floor.

2.1.2 Nature of cleaning task

In addition to the environmental constraints, the nature or type of cleaning the robot needs to perform must be analyzed, Since the robot is expected to moist clean, the robot needs to apply controlled amount of water to remove dirt from the floor surface. Unlike dry cleaning, wet cleaning required integration of water dispensing system, scrubbing and mopping unit as well as mild suction or drying mechanism. As per the demand the robot must clean the entire floor area at least once per cycle properly, this requires an efficient route planning, real time path correction in case of unexpected obstacles and strategy for covering all the area efficiently the cleaning efficiency may also be enhanced using features like dirt detection system.

2.1.3 Autonomy and System Expectations

The robot is expected to operate fully autonomous from start of cleaning cycle to its completion without human intervention, while navigating the hall avoiding obstacles or collisions, maintaining internal function such as battery and water usage monitoring, automatically returning to docking station in case of low battery or insufficient water. The robot system should include dynamic or pre-configured mapping using SLAM since the path can occasionally change due to presence of obstacles.

Further desirable features include auto battery saver mode, programmable cleaning zones, remote manual override, app-based control. Advanced control function such as geo fencing, voice assistant integration and multimode cleaning strategies were wished to be included by end user for higher adaptability and user interaction.

2.1.4 End-user Demands and Wishes

End-user expectations play a crucial role in shaping the design which is why the demands and wishes were analyzed to determine the requirements needed. Compact modular design with recyclable and reusable components and safety features such as collision avoidance, emergency stop was expected. Additionally simple maintenance and repair friendly architecture along with feature such as child lock, bump detection, warning signals and fire safety were wished. The list of demands and wishes in Appendix help distinguish between mandatory requirement(demands) and optional performance and enhancing features(wishes).

After analyzing and determining the requirements, they were further categorized into and technical, functional, operational and safety requirements which are explained below.

2.2 Technical Requirements

The technical requirements are the physical and structural requirements, power system specifications and modular design aspects necessary for the robot to function effectively and reliably within the LMS test hall. A compact and modular design is essential to ensure the robot can maneuver properly while cleaning all corners of the hall. The working width was decided to be 80cm based on the demands which provides optimal cleaning coverage while giving proper spatial maneuverability. Although a height of 10cm was wished it was determined that the height was inadequate to fit all the necessary component and have an optimal water tank volume and dust collector to clean the hall completely, therefore a height of 15cm was taken enabling access under the equipment in the hall while making it spacious enough for the individual parts of the robot. The parts such as scrubbing units and mopping units were designed as such they could be easily replaced. A eco-friendly rechargeable Lithium-Iron-phosphate battery was placed so that it could be replaced easily to ensure an energy efficient energy system. Tradeoffs between charging speed and capacity was considered and energy saving feature such Auto battery saver mode were considered to be added. Rigid and stable structure design along with easy to assemble and reparable part is required. This technical specification ensures the robots longevity, better performance and maintainability.

2.3 Functional Requirements

The functional requirement is the core requirement which include the cleaning, navigation and controlling system that are to be built in the robot for it perform its task effectively. The primary function of moist cleaning requires a vacuuming mechanism and integrated wet cleaning system comprising of water tank, scrubbing unit and mopping unit. The robot needs to be capable of autonomous operation throughout the cleaning cycle from starting, navigating, cleaning to returning to docking station. For advanced autonomy different sensor such LIDAR, optical

proximity sensor along with Raspberry Pi as controller were used. For operational flexibility, programmable cleaning zones, manual override via remote control and app based remote operation for user interaction can be added. Additional function such as dirt detection, self-cleaning, geofencing, voice assistance and multi cleaning modes are optional functional requirements that can enhance the adaptability and performance of the robot. These requirements directly influence the efficiency of the robot to perform the task it was designed for.

2.4 Operational Requirement

The operation requirement is the additional requirement based on end-user expectation regarding the usability and maintenance of robot during daily use. Since the robot is designed to clean a lab, it should provide safe and seamless operation in regards to the lab equipment as well as the people. This includes avoiding obstacle and collisions. For sustainability, it should incorporate recyclable materials and components with long term reusability. Eco-friendly cleaning agent as well as battery was determined to ensure eco-friendly robot. These requirements ensures that the robot is ecofriendly and also meet the modern standard for usability 2while being user friendly.

2.5 Safety Requirements

Safety is a critical prerequisite in the development of autonomous cleaning robot, particularly when operating in an indoor laboratory environment where delicate equipment and human interaction is prevalent. The robots must be designed to avoid collisions with people, equipment, machinery and structural elements through integrating of LIDAR, proximity sensors, camera and real time obstacle detection systems. In accordance with DIN EN ISO 12100, a structured risk assessment must be performed during the design phase to identify and minimize potential hazards. Functional Safety of the control system must comply with DIN EN ISO 13849-1 and DIN EN 61508, ensuring the critical safety function such as emergency stop, obstacle detection and docking etc. are reliable and fault tolerant. This includes an easily accessible emergency stop button as well as other optional requirement such as child lock, battery monitoring etc.

In term of electrical safety, the robot must be designed to follow DIN EN 60335-1 (VEDE 0700-1) which defines the requirement for household electrical appliances and similar appliances and DIN EN 62233 (VDE-0848-2233) to limit human exposure to electromagnetic fields. Additionally, as the robot operates wirelessly and autonomously, specific attention was given to product safety standard relevant to cleaning robots, particularly DIN EN IEC 63327 (VDE 0700-327) for robotic battery powered electrical cleaning appliance used in commercial settings. These frameworks ensure the robot is safe for continuous operation in workspaces, minimize electrical and mechanical risks, and maintain compliance with essential safety norm for autonomous systems.

3 Conceptual Design

The conceptual design phase involves the development of structural and functional layout of autonomous cleaning robot, integrating the critical components to meet the above mentioned technical, functional, operational and safety requirements. The model designed below represent a compact, modular, and efficient configuration tailored for the LMS test hall.

3.1 General Layout and Structure

The robot is designed to have compact and modular geometry with optimal maneuverability, source of energy, required component for cleaning, controlling and navigation. The design emphasizes ease of assembly, material recyclability, and compliance with relevant safety standards. The robot can be divided into distinct parts based on the functionality as follows:

1. Main body
2. Motion Unit
3. Cleaning Unit
4. Energy Unit
5. Control and Navigation Unit

3.1.1 Main body

The main body serves as the structural backbone of the robot. It is a square shaped structure of 50cm x 50cm x 20cm that houses all the necessary component such water tank, vacuuming unit, battery pack, controllers and sensors etc. The body is designed to be rigid, stable and compact allowing the robot to navigate confined spaces while ensuring mechanical integrity. Its modular configuration allows for quick assembly and component replacement.

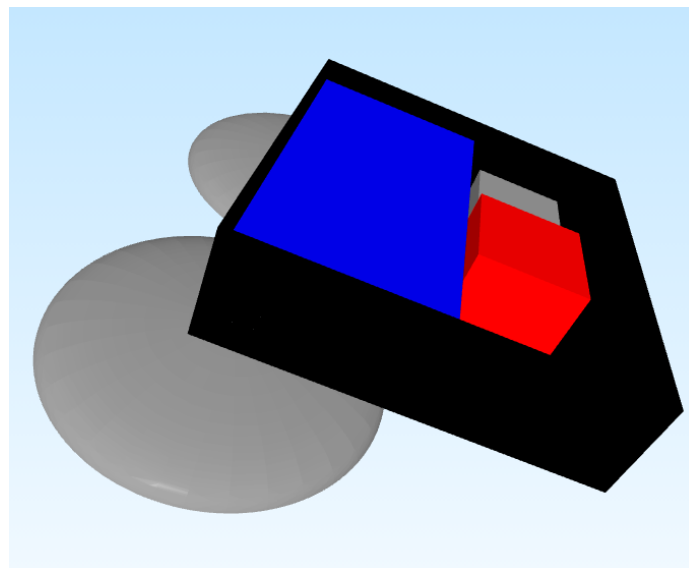


Figure 4.1: Main body of the robot housing the components

3.1.2 Motion Unit

The Motion Unit is responsible for enabling the autonomous cleaning robot to navigate the LMS test hall with precision and stability. It consists of a differential drive system with two independently driven rear wheels and a passive omni-directional support wheel to maintain balance and smooth maneuvering.

The two rear drive wheels, each 8 cm in diameter, are positioned between the vacuum nozzle and mopping unit, optimizing weight distribution and cleaning path efficiency. These wheels are powered by Pololu 37D (50:1) 12V DC gear motors, each equipped with an integrated encoder for real-time odometry and closed-loop speed control. The encoder feedback supports accurate motion tracking, crucial for mapping, localization, and autonomous path planning.

Motor Specification:

Model: Pololu 37D (50:1) with Encoder

Type: 12V DC Gear Motor

Speed: 200 RPM

Torque: 21 kg·cm

Weight: 205 g per motor



Figure 4.2: Motor for the wheels

This motor configuration offers a good balance between torque and speed, ensuring the robot can traverse the indoor test environment efficiently while maintaining sufficient power for obstacle negotiation and directional control. The inclusion of a passive omni-directional wheel at the front allows for smooth rotational movements and stability on flat concrete surfaces

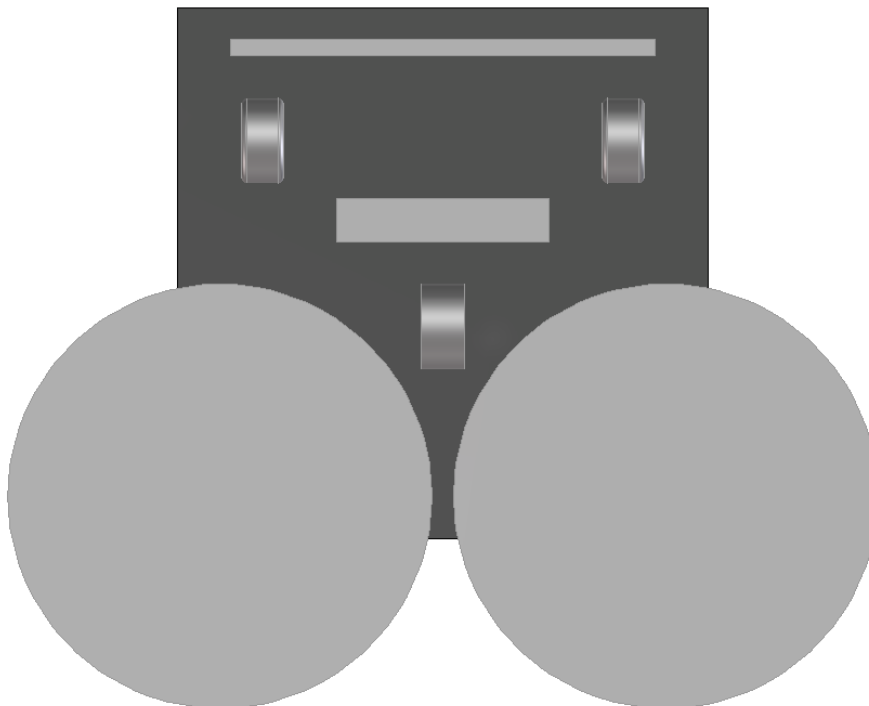


Figure 4.3: Bottom view of robot illustrating the position of wheels

3.1.3 Cleaning Unit

The cleaning unit is responsible for the robot's core cleaning functionality, it consisted of Vacuum system, scrubbing unit and mopping unit along with a water tank for the storage of water for wet cleaning as well as a storage unit for the vacuumed dirt collection. These unit were carefully positioned to ensure the stability and maximum efficiency.

The scrubbing unit was a round brush structure at the front of the robot as illustrated in Figure 4.4 with a diameter of 40cm with 16cm over the edge of the main body as illustrated in Figure 4.3 making the working width of 80cm as per demand. The same Polulu 37D motor was also taken for both scrubbing unit without encoder. The mopping unit was at the rear end with a 10cm width and 40cm long.



Figure 4.4: Example of a scrubbing and mopping unit

The vacuum was at mid-section behind the scrubbing unit for suction of dirt which was collected using a vacuum system illustrated in Figure 4.5 and Wderair Brushless Blower with 12V, 1A and 3000RPM speed for suction illustrated in Figure 4.6.

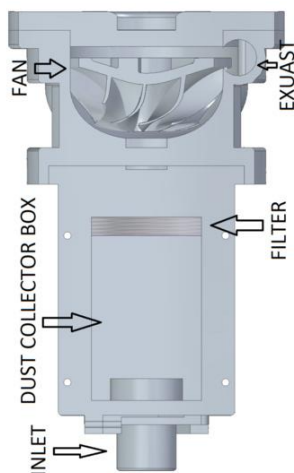


Figure 4.5: Vacuum system



Figure 4.6: Motor for Vacuum

The water tank of volume 13L with the dimension of 24cm x 44cm x 12 cm was placed at the front followed by container for dirt collection of dimensions 15cm x 15cm x 12cm as illustrated in the Figure 4.7.

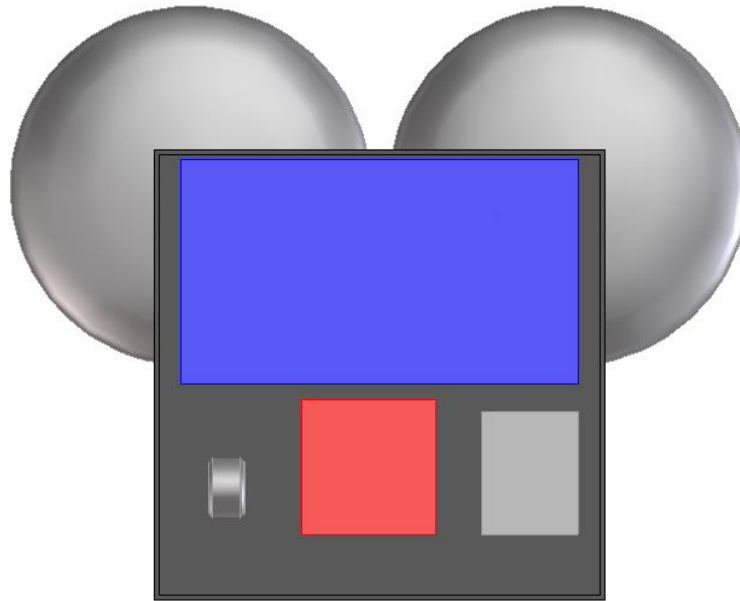


Figure 4.7: Top view showing the placement of Water tank (Blue) and Dirt container (Red)

A Fafeicy G928 12V peristaltic pump with a flow rate of 500ml /min as illustrated in figure 4.8 was used in the water tank to supply water to the mopping unit.

DC 12V Brush Gear Motor



500ML/min Flow Rate

Figure 4.8: Water pump

3.1.4 Energy Unit

The energy unit comprised of Battery that provided power required by different component to function properly. A Lithium Iron Phosphate (LiFePO₄) Battery of 12V 15Ah battery was used for the consistent and reliable power supply with no EMI/RFI interference, ensuring quiet and efficient operation. This battery is inbuilt with Protection circuit module (PCM) ensuring internal cell balancing and comprehensive protection against overcurrent, undervoltage, overvoltage, and short-circuiting. The battery has extended battery lifespan with up to 2000 charge cycles,

Additionally, LiFePO_4 batteries are free from harmful chemicals like lead and acid, making them an eco-friendly choice ensuring environmental sustainability. An illustration of battery is shown in Figure 4.9.



Figure 4.9: LiFePO_4 Battery

3.1.5 Control and Navigation Unit

The control unit comprises the onboard microcontroller responsible for managing and coordinating the robot's overall functionality. In this design, a Raspberry Pi is used as the master controller to oversee and control all components of the robot, ensuring smooth operation and efficient task scheduling.

The navigation unit consists of a combination of sensors that enable navigation and environmental mapping. These include a 3D LiDAR sensor (LD06), proximity sensors, and a camera. Together, they facilitate real-time obstacle detection, path planning, and autonomous return to the charging dock. Mapping is achieved through the integration of the 3D LiDAR sensor and the camera, which enhances the robot's adaptability to changes in its surroundings.

The proposed LD06 LiDAR sensor incorporates Time-of-Flight (TOF) technology to measure distances based on the flight time of laser pulses, allowing detection within a range of up to 12 meters. It uses an advanced algorithm to provide 360-degree environment scanning, autonomous mapping, and obstacle detection. The sensor supports high-speed wireless data transmission and is equipped with brushless motors, offering a lifetime of up to 10,000 hours.

The LD06 is suitable for the robot since it delivers high-frequency and high-precision distance measurements and accurate environmental mapping. Additionally, it is fully compatible with Raspberry Pi, ensuring seamless integration with the robot's control system.

Additionally, a camera was used on front of robot to enhance object detection, navigation and mapping as well as 3 proximity sensor two on sides and one on front was ensure the robot detect the changes in surface and detect presence of nearby objects. These sensors are illustrated in Figure 4.10.



(a)



(b)

Figure 4.10: LiDAR sensor(a) and Proximity sensor(b)

To further enhance the robot's autonomy and intelligence, the system will integrate the Robot Operating System (ROS) as a middleware framework. ROS facilitates efficient sensor integration, modular software development, and real-time data communication between components. Additionally, an AI-based object detection model such as YOLO (You Only Look Once) will be employed to enable advanced visual recognition tasks. This includes detecting and classifying objects like dirt and obstacle in real time, improving the robot's ability to interact with its environment and make intelligent navigation decisions based on visual input.

4 Route Planning

The robot is designed to methodically cover the entire floor area of LMS test hall using a predefined route. The route planning strategy prioritizes efficiency and simplicity, minimizing overlaps while ensuring no section is missed. The predefined route for one complete cleaning cycle is illustrated in Figure 4.11.



Figure 4.11: Predefined route of robot from start to finish for a cleaning cycle

A calculation was performed to estimate the time required for the robot to clean the designated area. The total cleaning area consists of two rectangular sections and an additional open space, calculated as follows:

$$\text{Total area} = 2 \times [(75\text{m} - 6\text{m}) \times 3\text{m}] + (20\text{m} \times 6\text{m}) = 414\text{m}^2 + 120\text{m} = 534\text{m}^2$$

$$\text{Speed of the wheel} = \text{circumference of wheel} \times \text{RPM} = 0,2513\text{m} \times 200 = 50.26\text{m/min}$$

$$\text{Area navigated per minute} = \text{cleaning width} \times \text{speed} = 0.8\text{m} \times 50.26\text{m/min} = 40.2\text{m}^2/\text{min}$$

$$\text{Total time} = \frac{\text{Total area}}{\text{Area cleaned per minute}} = \frac{534\text{m}^2}{40.2\text{m}^2/\text{min}} = 13.28\text{ mins}$$

However, this calculation is based on ideal conditions and assumes navigation without any delays. In reality, cleaning typically requires slower movement than basic navigation due to mechanical constraints and the presence of obstacles. Therefore, if the cleaning speed is approximately half of the navigation speed, the estimated total cleaning time would realistically range between 20 to 25 minutes.

5 References

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6 Appendix

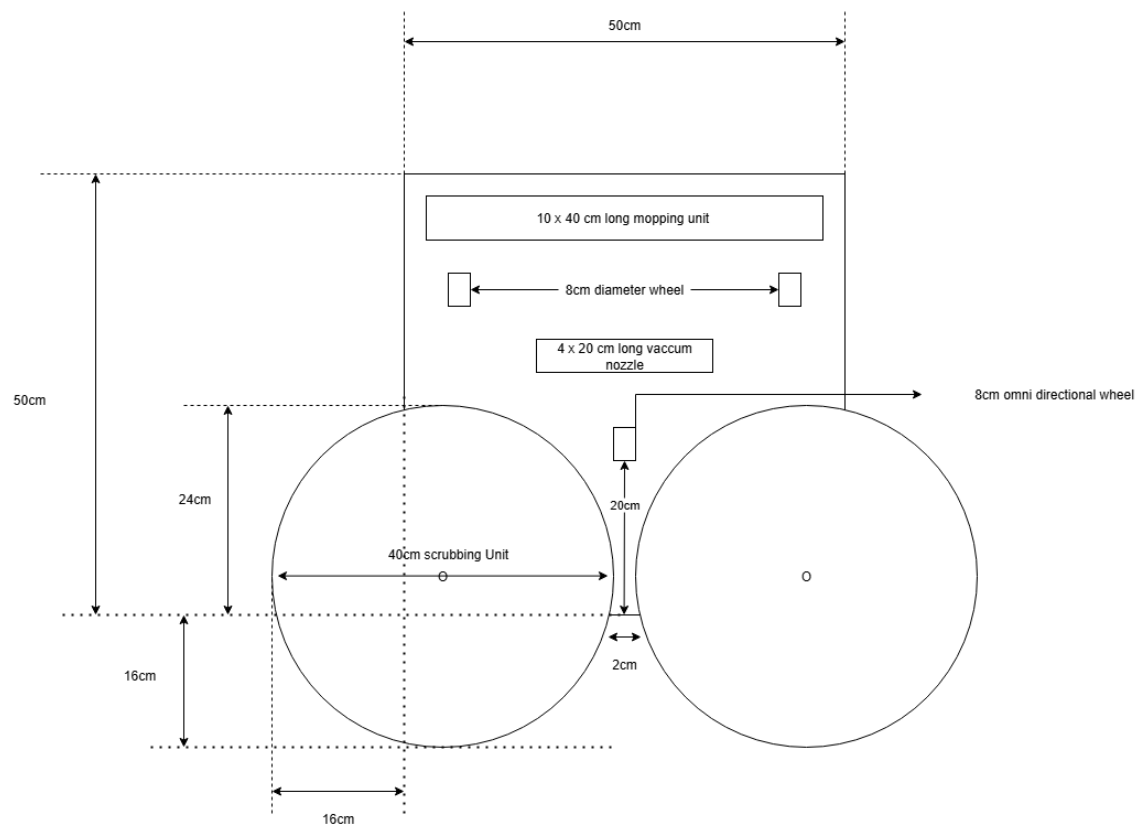
6.1 Autonomous Cleaning Robot Requirements

Number	Requirement group	Art	Features
1	Geometry	Demand	Compact design
2	Geometry	Demand	Modular design (cleaning stuffs such as brush or filters can be easily replaceable)
3	Geometry	Wish	Less height (probably less or equal to 10 cm)
4	Geometry	Demand	Working width 80 cm
5	Geometry	Demand	Small so it can maneuver every corner properly
6	Geometry	Wish	3D model, simulations
7	Energy	Demand	Rechargeable Battery
8	Energy	Demand	Faster charging, lower capacity or slower charging, bigger capacity
9	Energy	Demand	Energy consumption not too high (Efficiency)
10	Energy	Demand	Electrical-mechanical energy conversion
11	Energy	Demand	Compatibility with building energy sources
12	Energy	Demand	Replaceable battery
13	Energy	Wish	Auto Battery Saver mode when not in use
14	Function Working Function	Demand	Wet cleaning system with water tank and scrubbing unit
15	Function Working Function	Wish	Detection of dirtier areas

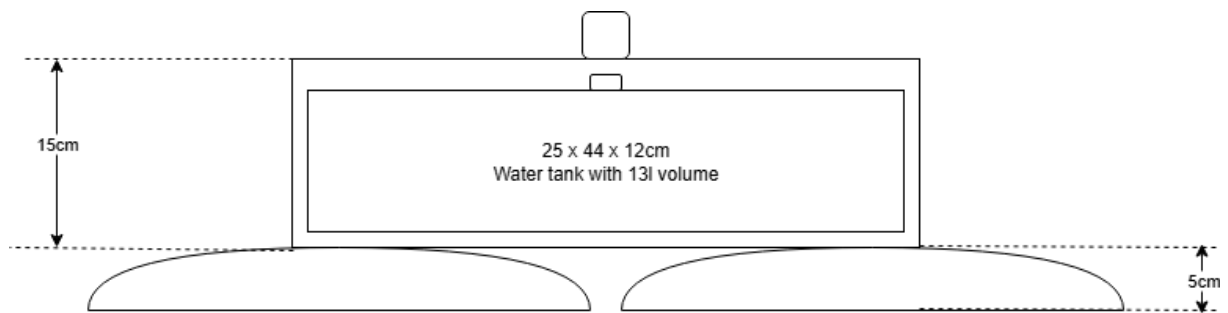
16	Function Working Function	-	Demand	Vacuuming
17	Function Working Function	-	Demand	Self return to charging dock during low battery warning
18	Function Working Function	-	Wish	Self cleaning
19	Function Working Function	-	Wish	Multi cleaning modes (zigzag, straight line or fast cleaning, deep cleaning)
20	Function Working Function	-	Wish	Optimisation for specific surface
21	Function Working Function	-	Wish	Cleaning efficiency metric (% of cleaned floor per m ² defining speed of machine and time)
22	Controlling		Demand	Programmable cleaning zone
23	Controlling		Demand	Manual override option via remote control
24	Controlling		Demand	App control (Remote)
25	Controlling		Wish	Geofencing
26	Controlling		Wish	Autonomous Navigation (LIDAR)
27	Controlling		Wish	Voice assistance (Alexa, Siri, Gemini)
28	Recycling		Demand	Recyclable Material (reusable material such as steel, aluminum etc.)
29	Recycling		Demand	Reusable filters and washable brushes
30	Recycling		Wish	Compatible with environmentally friendly cleaning agents
31	Recycling		Wish	Battery infrastructure with few rare earth metals

32	Safety	Demand	Collision avoidance
33	Safety	Demand	Emergency stop button
34	Safety	Demand	Safe for human interaction and delicate lab equipment
35	Safety	Demand	Sensors and cameras for safety
36	Safety	Wish	Warning wizard (with light signal or buzzer)
37	Safety	Wish	Child lock and pet safety
38	Safety	Wish	Bump detection
39	Safety	Wish	Fire safety by monitoring battery temperature
40	Safety	Wish	Navigation and obstacle avoidance (Proximity Sensors)
41	Safety	Wish	Biodegradable cleaning agents
42	Assembly & Production	Demand	Easy to assemble modular parts
43	Assembly & Production	Demand	Rigid and stable structure
44	Assembly & Production	Wish	Easy to repair

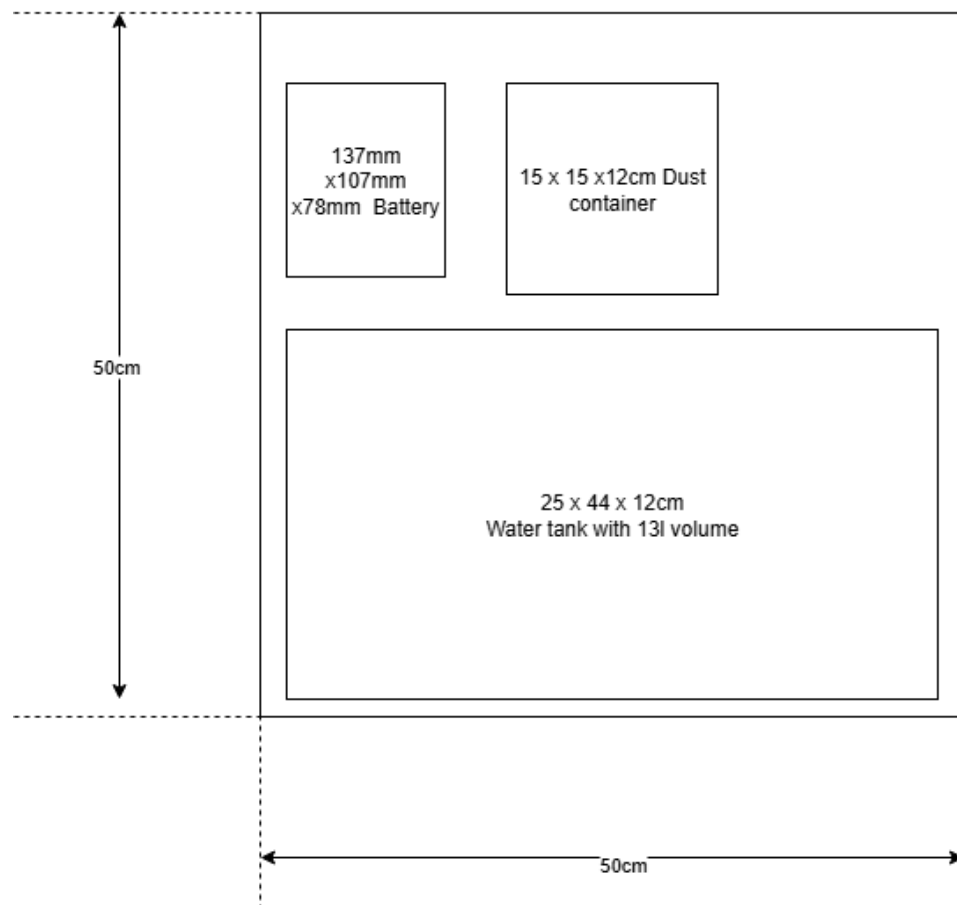
6.2 CAD 2D Designs



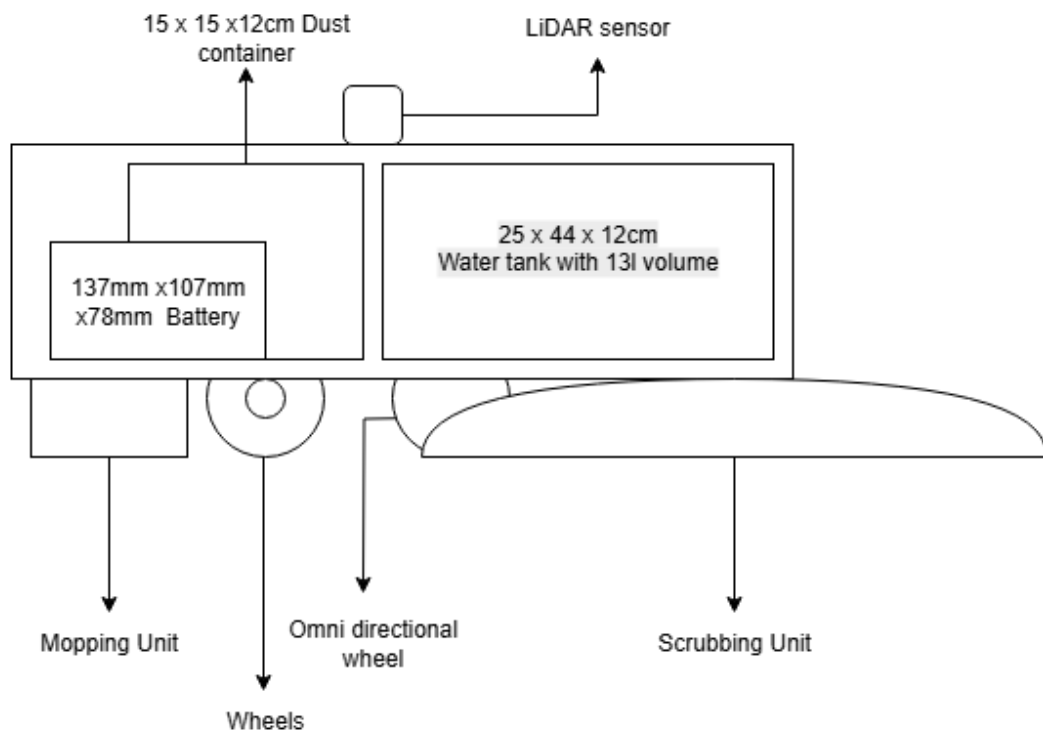
Bottom view



Front view

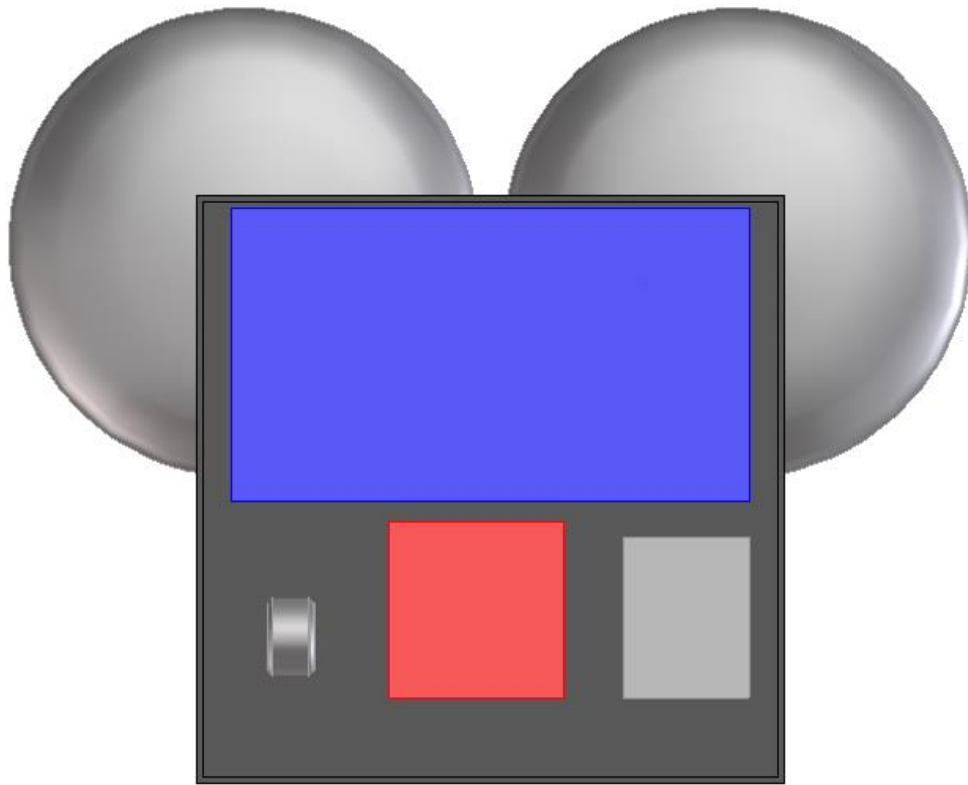


Top view

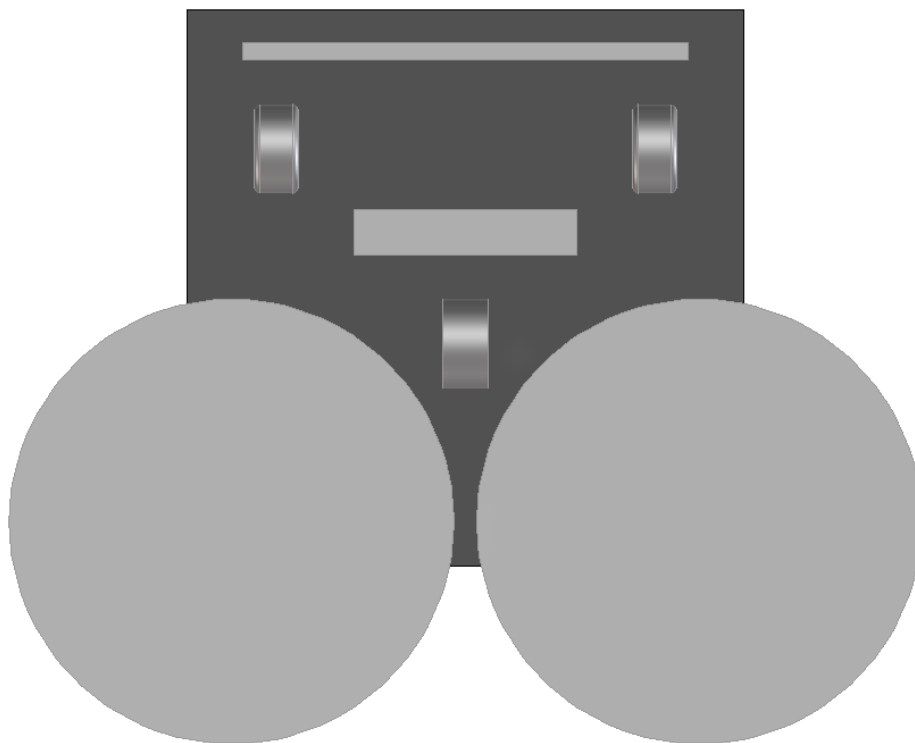


Side view

6.3 3D CAD Designs



Top View



Bottom View